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Mohammad Mahdi Malayeri

Professional Profile

Researcher in data mining, machine learning, and network optimization, seeking a PhD position. My MSc research applied pattern recognition to traffic data to reduce spectrum fragmentation in elastic optical networks, using OPNET simulation and decision-tree models. I currently teach computer science and photovoltaic system analysis (PVSyst) at the Technical and Vocational University, and lead data analysis and business intelligence for a family agricultural business, where data-driven crop planning cut water consumption by nearly 40%. I bring strong hands-on experience in Python, data analytics, computer hardware, and network troubleshooting.

Education

University of Qom — Qom, Iran
MSc in Information Technology Engineering (E-Commerce) — Sep 2017 – Jan 2022

- Thesis: "Optimization of Fragmented Spectrum in Elastic Optical Networks Using Pattern Recognition"
- Supervisor: Dr Mohsen Nickray
- Analyzed real-world IPTV and VoD traffic to find recurring packet patterns between node pairs, and used a decision-tree model to predict frequency-slot demand so spectrum could be reserved in advance
- Combined Push-Pull defragmentation with a greedy route-grouping algorithm; evaluated through OPNET simulation on the 14-node NSFNET topology

Islamic Azad University — Semnan, Iran
BSc in Computer Software Engineering — Sep 2012 – Aug 2017

- Thesis: "Introduction to Point-to-Point Radio Networks"
- Supervisor: Dr Mohammad Karim Sohrabi

Research Interests

- Elastic Optical Networks, Spectrum Allocation & Defragmentation, Machine Learning, Data Mining, Agricultural Data Analytics, Business Intelligence, Renewable Energy & Photovoltaic Systems

Research Projects

- **NSFNET Topology Simulator for Elastic Optical Networks** (OPNET, C++). Simulation model of the NSFNET topology with custom node, packet and process models that generate connection requests with Poisson arrivals and periodic traffic. GitHub
- **Real-time Packet Pattern Finder** (C++, Python). Stream-based tool that groups requests by node pair while the simulation runs, detects periodic requests, and tracks frequency-slot demand over time. GitHub
- **AI-Based Dynamic Resource Allocation** (Python, scikit-learn, Stable-Baselines3). Random Forest prediction of connection holding times and a reinforcement-learning (DQN) prototype for frequency-slot allocation. GitHub
- **Data-Driven Crop Planning and Water Optimization** (Python, Excel, Power BI). Year-by-year analysis of farm data for alfalfa, corn, wheat, and barley that cut water consumption by nearly 40% while increasing production.

Academic Experience

Technical and Vocational University — Tehran, Iran
Adjunct Professor of Computer Science (part-time) — Sep 2025 – Present

- Teach computer science and photovoltaic system analysis (PVSyst), combining IT education with renewable energy applications

Ministry of Education — Varamin, Iran
Computer Science Teacher (Conscription Service) — Jun 2022 – Jun 2024

- Taught computer science to 12th-grade technical high school students, focusing on practical IT skills and programming fundamentals
- Developed lesson plans, supervised projects, and prepared students for national vocational exams
- Awarded Technical and Vocational High School Teacher of the Year (2023–2024)

Islamic Azad University Semnan, Iran
Teaching Assistant 2014 – 2017

– Computer Architecture, Logic Circuits, and Microprocessors; held problem-solving practice classes

Industry Experience

Malayeri Agro-Industry Varamin, Iran
Managing Director Apr 2017 – Present

– Lead data analysis and business intelligence for a family-owned agricultural and dairy farming business
– Collect, clean, and analyze operational and production data on crops, livestock, herd performance, and resource use
– Developed year-by-year, data-driven crop planning that cut water consumption by nearly 40% while increasing production

Dibasabz Tehran, Iran
Information Technology Consultant (part-time) Mar 2019 – Oct 2020

– IT consulting and technical support for e-commerce platforms (WordPress, WooCommerce), focusing on SEO and server performance
– Set up, configured, and maintained Windows Server environments for reliable hosting and data security

Mahad Rayane Varamin, Iran
Personal Computer Support Technician Jan 2014 – Feb 2016

– Hardware installation, assembly, troubleshooting, and repair; on-site support for individuals and small businesses

Babbage Rayane Varamin, Iran
Trainee Apr 2012 – Dec 2013

– Computer hardware assembly, maintenance, and troubleshooting as part of a technical training program

Honors

- Technical and Vocational High School Teacher of the Year 2023–2024
  Ministry of Education, Iran
- Ranked 341st (top 3%) in the National MSc Entrance Exam for Information Technology 2017
  National Organization for Educational Testing, Iran

Skills

- **Programming Languages:** Python, C++
- **Data Science & Machine Learning:** pandas, NumPy, scikit-learn, Matplotlib, Reinforcement Learning
- **Analytics & BI:** Microsoft Power BI, Microsoft Excel
- **Simulation & Energy:** OPNET Modeler, PVSyst
- **IT & Systems:** PC hardware assembly, Windows configuration and troubleshooting, network troubleshooting, Windows Server
- **Web & E-commerce:** WordPress, WooCommerce, SEO
- **Tools:** Git, Microsoft Office

Languages

- **Persian:** Native
- **English:** Fluent

Professional Activities

- Member, Network Lab, University of Qom Sep 2018 – Oct 2020
  – Attended weekly meetings where recent networking papers were presented and discussed

Referees

Available upon request.